

# Swissgrid Single-Price Mechanism

## Analysis on System Imbalance and Intraday Trading

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**Confidential**

### Abstract

To operate electrical machines safely in Europe, the electrical grid should always run at 50 Hertz. This frequency is impacted by production and consumption. When production is higher than consumption, the frequency increases; otherwise, it decreases. Switzerland's Transmission System Operator (Swissgrid) uses ancillary services to keep this frequency as stable as possible. If the grid is not balanced, the TSO will automatically send signals to balance responsible parties (BRPs) to adjust their production accordingly. BRPs are thus financially compensated to balance the grid with their ancillary services offers. On the other hand, they are penalized when they unbalance the grid. To avoid paying penalties, open positions can be closed until 5 minutes before delivery on the intraday market. A common rule is to keep those positions as low as possible. Residual positions could still be exchanged between BRPs a couple of days later during the so-called Post Scheduling Adjustment (PSA). Under the prior dual-price system, positive and negative imbalances were settled at distinct prices. However, Swissgrid introduced a single-price imbalance settlement mechanism starting 1 January 2026. BRPs that are in the opposite side from Swissgrid and therefore help to balance the grid are now compensated. This creates stronger financial incentives for BRPs to trade in the direction that restores system equilibrium. This report presents a quantitative assessment of the mechanism's early impact on imbalances and prices.

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# 1. Introduction

The main goal of every BRPs was to keep their positions as balanced as possible to reduce their power balancing costs. With the old dual-price mechanism for balancing energy, BRPs paid penalties whether they were short or long for each hour of a specific day. To minimize those costs, they could exchange positions with other BRPs at the spot price during the PSA. For example, if BRP A was long 5 MW for a specific hour and BRP B was 5 MW short for the same hour, they could exchange this position. Both BRPs were balanced and did not need to pay any penalties for that hour. From a TSO perspective, the goal is to balance the grid and keep it as stable as possible to avoid any major issues. The increase of renewable energies connected to the grid can affect this stability. Production forecasts can fluctuate a lot and put a lot of pressure on the grid. This is particularly true for Sunday and Monday, where forecasts are sometimes not amended during the weekend. When the forecast from a balance group tends to underestimate or overestimate production, this is also often the case for other BRPs. It increases the imbalance on the grid and could potentially lead to dangerous situations. From 2026, the new single-price method is used to reduce the need for control energy. According to Swissgrid<sup>1</sup>, this new mechanism creates financial incentives for all BRPs to contribute to the system equilibrium. Keeping open positions in the opposite side of Swissgrid will be financially compensated. Consider this example: if Swissgrid is short and BRP A is long for the same hour, BRP A will be compensated and will have no incentive to exchange its position with BRP B, which is short. BRP B will therefore pay penalties. The PSA is therefore no longer relevant and no longer used. It is sometimes not easy for a BRP to know its exact intraday position due to poor forecasts. It is even harder to predict the exact position of Swissgrid for the next 60 minutes. However, if one can estimate those two positions in a relatively precise way, it can be tempting to take specific positions, either on the intraday market, by keeping already well-known positions, or by activating more energy from mFRR (manual Frequency Restoration Reserve). However, a risk remains: if several BRP deliberately take positions in a certain direction, there is a probability that Swissgrid will shift in the same direction. Those BRPs will then pay penalties. It is therefore important to monitor the profits and losses of such activity.

## 1.1 Project Objectives

The initial goal of this study is to determine whether the new single-price mechanism gives enough incentives to BRPs to improve grid stability by taking specific intraday positions, and to assess the impact on balancing energy prices. If this is true, the imbalances on the grid should decrease, as well as prices, because activated offers are selected according to a merit order mechanism. To answer this question, imbalances and imbalance prices before and after 2026 are analysed. More specifically, the study analyses the average imbalance per 15min for January, February, March and April in 2025 and compares them to the results for 2026. The same applies for the imbalance prices. This is done by analysing historical data provided by Swissgrid. A secondary analysis evaluates the economic viability of an intraday strategy that leverages imbalance signals to optimise near gate-closure trades against Swissgrid's settlement prices.

## 1.2 Imbalance and Price Calculations

To keep the grid stable, Swissgrid first needs to forecast production and consumption using advanced models that consider several features such as historical data and weather conditions. When the frequency of the grid needs to be restored, ancillary services are used. There are three types of frequency control: primary control (FCR), secondary control (aFRR) and tertiary control (mFRR). Primary control is used within seconds, followed by secondary control after 30 seconds, which is eventually replaced by tertiary control after 15 minutes if needed. Exports and imports at the Swiss border are also possible. Therefore, the total imbalance of the grid per 15 minute interval equals the sum of the activations of FCR, aFRR, mFRR and transfers at the border. The imbalance price depends on the price of activated energy control, which can be the maximum VWAP between aFRR and mFRR when the grid is short, or the

minimum VWAP between aFRR and mFRR when the grid is long. Under the new single-price regime, a unique price is applied regardless of the direction of imbalance, replacing the former dual price structure.

## 2. Data and Methodology

### 2.1 Data Sources

Three datasets underpin the analysis.

- `control-area-balance-2025.csv`  
Full-year 2025 Swissgrid<sup>2</sup> control area balance, 15-minute resolution. Covers imbalance volumes (in MW), aFRR, mFRR, RR activations, FRCE, NRV flows, and the dual price columns (long, short, and the 2026-equivalent retroactive price).
- `control-area-balance-2026.csv`  
January–April 2026, 15-minute resolution. Columns mirror the 2025 structure but the dual-price fields are replaced by a single price field, reflecting the new settlement regime.
- `Trades_jan_apr.csv`  
A sample of intraday trades executed on the Swiss and German intraday market, January–April 2026. Includes trade direction (buy/sell), volume and price.

Datetime alignment: the 2026 CSV timestamps are provided in UTC and are converted to UTC+01:00 (Swiss winter time) before merging with 2025 data, ensuring consistent comparison across both years.

### 2.2 Analytical Approach

- Monthly imbalance aggregation: for each month (January - April), 15-minute observations are grouped and the mean positive and negative imbalances are computed separately, enabling sign-asymmetric analysis.
- Day-group segmentation: the dataset is split into Tuesday - Saturday (standard business schedule) and Sunday–Monday (weekend) sub-panels to test whether the single-price effect is uniform across day types.
- Imbalance persistence analysis: a forward-looking probability is computed. For each imbalance threshold level, what fraction of events at or above that threshold maintained the same imbalance direction 1.5 hours later (2025) or 1 hour later (2026)
- Statistical testing: the Mann-Whitney U test (non-parametric, no normality assumption) is applied to compare absolute imbalance distributions between 2025 and 2026, at the overall level and separately for positive and negative episodes.
- Price comparison: the 2025 dataset contains a retroactive price per 2026 column, allowing a like-for-like price comparison using the new pricing formula applied to both years.
- Back-testing: the intraday trades are matched to the 2026 hourly imbalance price (average of four 15-minute observations per hour). A threshold rule triggers re-routing to Swissgrid settlement instead of intraday settlement when the prior-hour imbalance exceeds an absolute threshold and the trade direction is in the opposite direction.

### 3. System Imbalance Analysis

#### 3.1 Monthly Imbalance — Overall (Jan - Apr 2025 vs 2026)

The chart below compares monthly average positive and negative imbalances across January - April for both years. The visual reduction in bar heights between 2025 and 2026 is immediately apparent for all four months and for both sign directions.

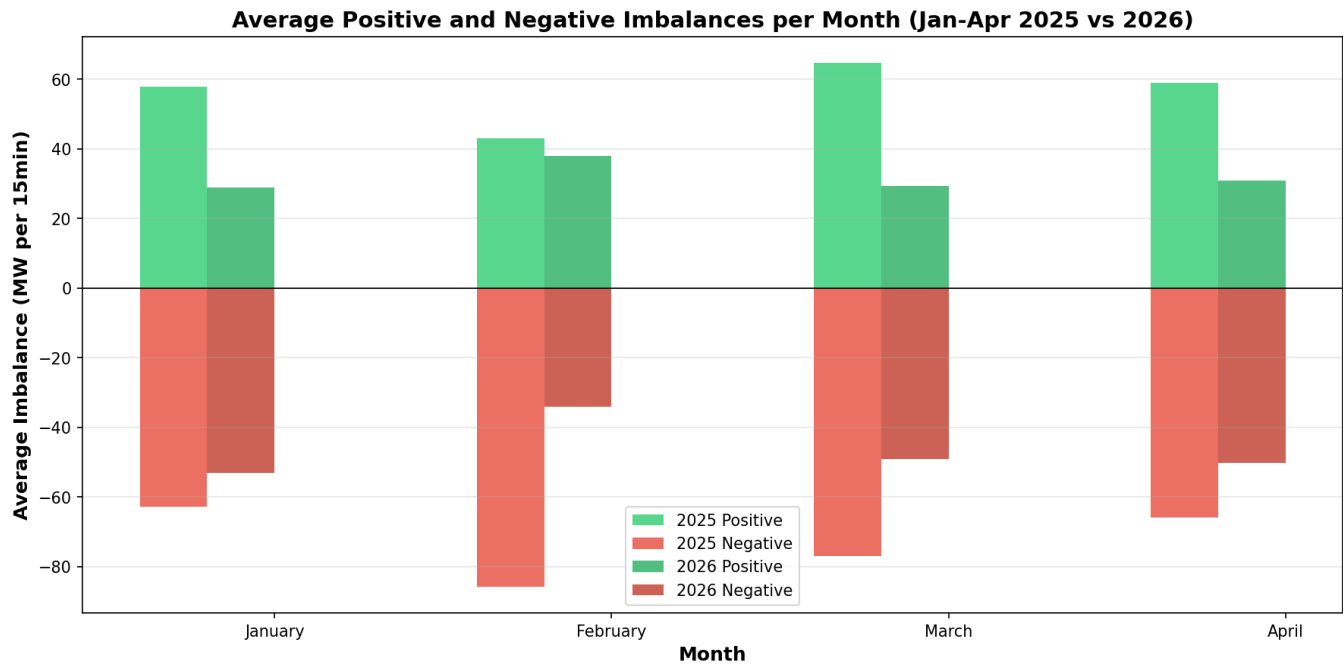


Figure 1 — Average positive (green) and negative (red) system imbalance per 15-minute interval, by month (Jan–Apr 2025 vs 2026).

January and February show the most pronounced positive imbalance reductions, consistent with the hypothesis that BRPs adapted their trading behaviour rapidly in the new regulatory environment at the start of the year. March and April maintain lower imbalance levels than the corresponding 2025 months, suggesting the improvement is structural rather than transient. The extreme-events also collapsed: 215 positive events above 400 MW in 2025 versus only 16 in 2026. Above 600 MW, 2026 registers zero in both directions.

#### 3.2 Monthly Imbalance - By Day Group

Splitting the data by day-of-week group reveals whether the aggregate improvement conceals asymmetries between working days and the weekend transition. This distinction is particularly important: as noted in the introduction, Sunday and Monday are days when forecast amendments are less frequent, historically contributing to higher imbalance levels.

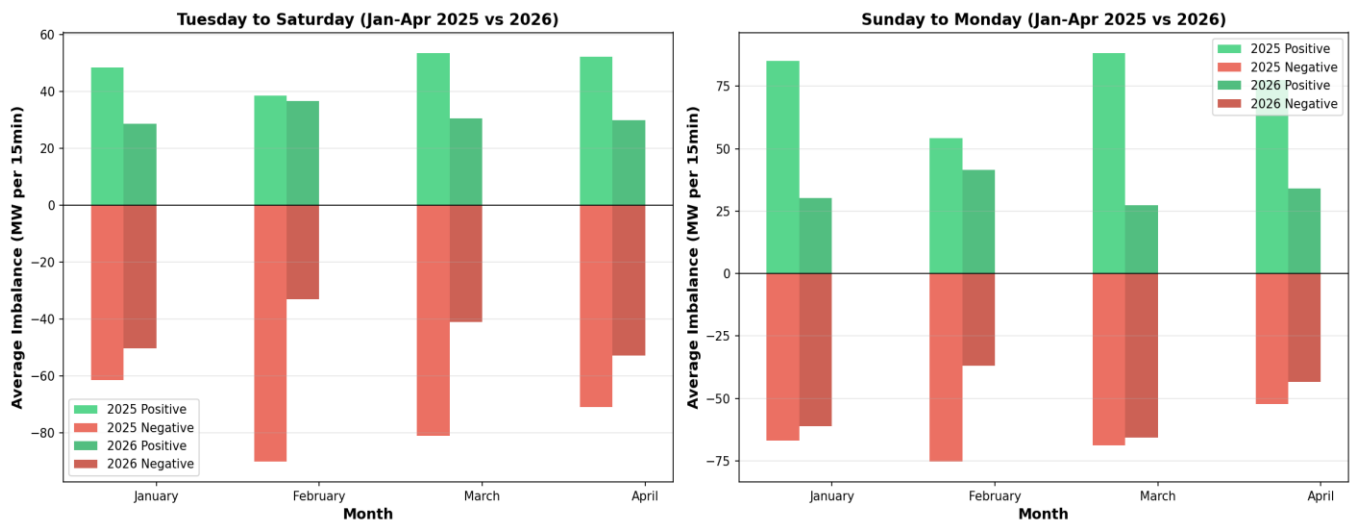


Figure 2 — Average positive and negative imbalances per 15-minute interval by month, segmented by day group: Tuesday–Saturday (left) and Sunday–Monday (right).

Both sub-panels demonstrate consistent improvement in 2026 relative to 2025. The Sunday - Monday group shows slightly higher average imbalances in both years confirming the weekend effect described in the introduction but the proportional reduction between years is comparable to the weekday panel.

## 4. Statistical Significance

The Mann-Whitney U test compares the full distributions of 15-minute absolute imbalance values between 2025 and 2026. This non-parametric test makes no assumption of normality, making it appropriate for the heavy-tailed, non-Gaussian distributions typical of power system imbalance data.

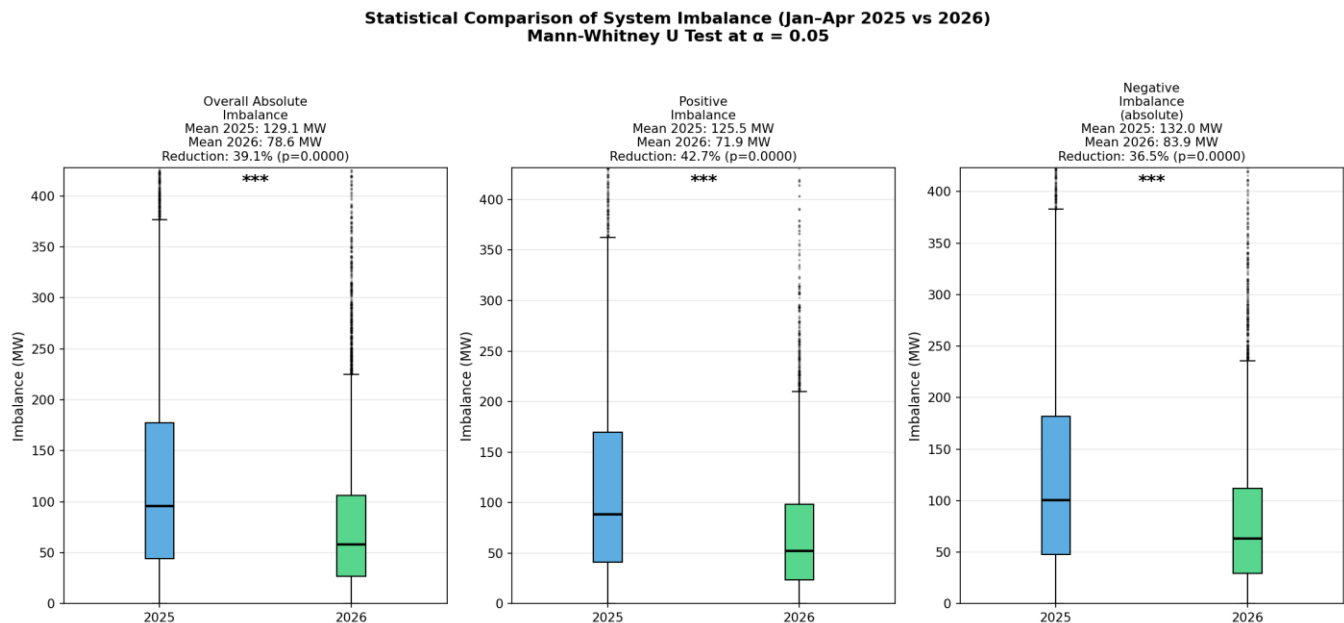


Figure 3 — Box plots of absolute imbalance distributions (January–April 2025 vs 2026) for overall, positive, and negative imbalances. \*\*\* denotes  $p < 0.001$ .

All five tests yield p-values well below the 0.001 threshold, providing overwhelming statistical evidence that the reduction in system imbalance is not attributable to random variation. The consistency across

imbalance sign and day-type classifications reinforces the conclusion that the single-price mechanism has had a broad systemic effect on BRPs behaviour.

## 5. Imbalance Price Analysis

### 5.1 Overall Monthly Prices

The chart below compares the average imbalance settlement price (EUR/MWh) for positive and negative imbalance episodes across January–April for both years, using price per 2026 column for the 2025 data to ensure methodological consistency.

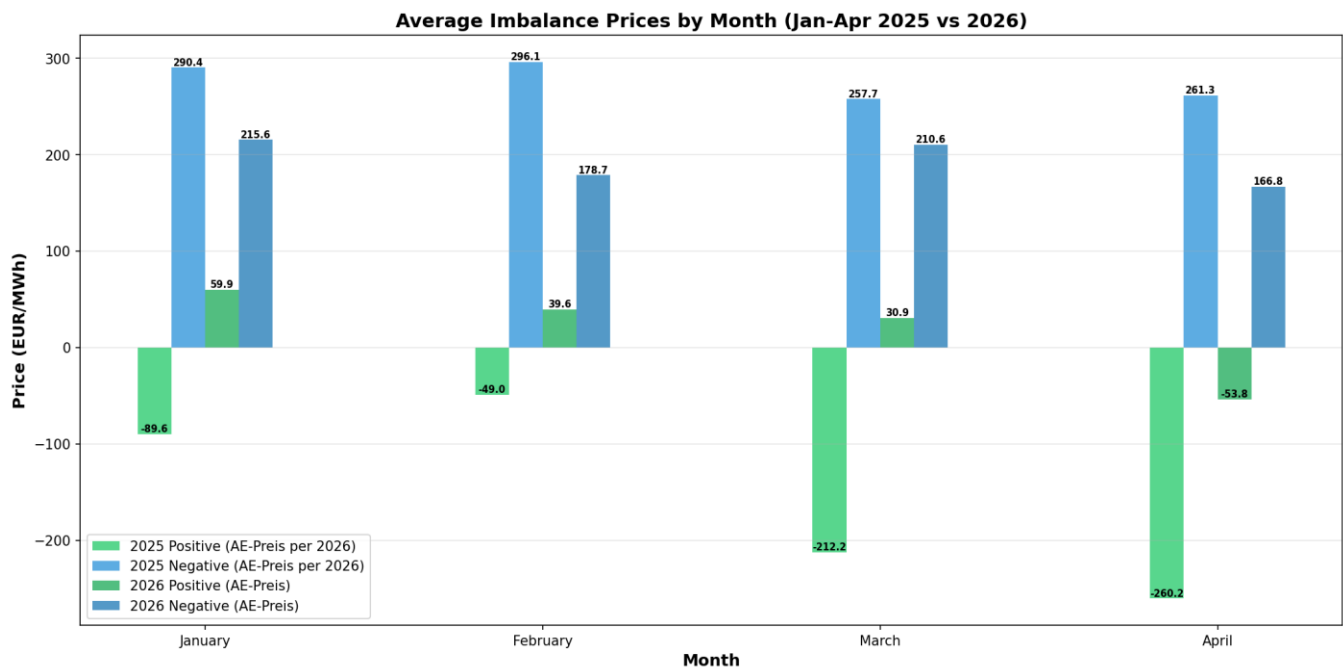


Figure 4 — Average imbalance settlement prices (EUR/MWh) by month and imbalance sign (Jan–Apr 2025 vs 2026). 2025 bars use the retroactive AE-Preis per 2026 formula.

Prices in 2026 are generally lower than their 2025 equivalents when re-calculated on the new single-price basis. This reflects one effect: smaller imbalance volumes require less balancing energy activation, reducing marginal costs.

## 5.2 Price Comparison by Day Group

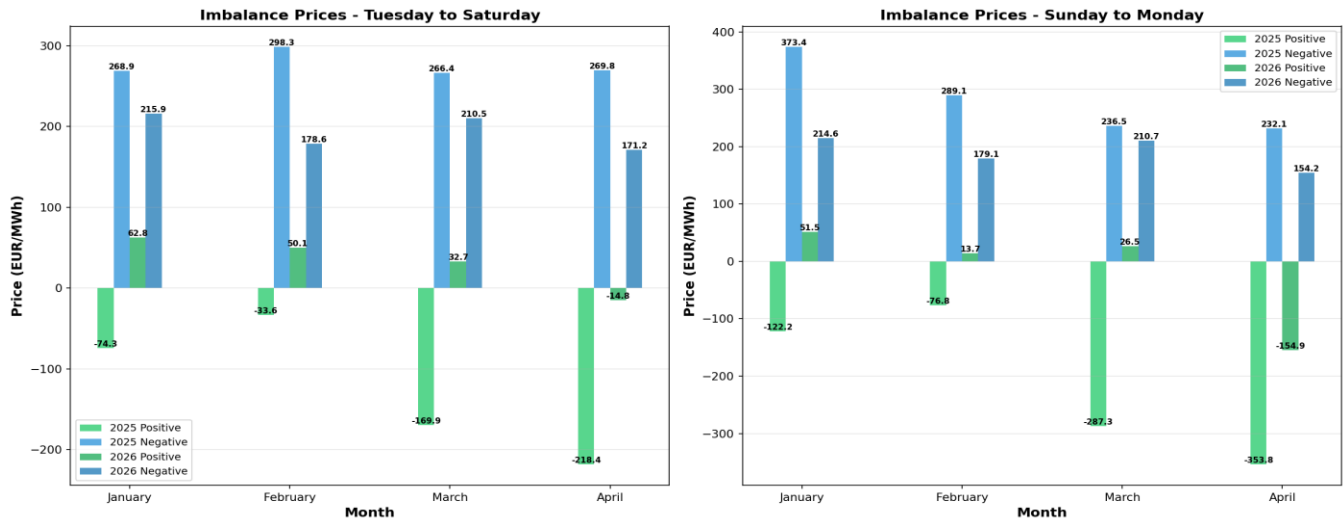


Figure 5 — Average imbalance prices (EUR/MWh) by month and imbalance sign, segmented by day group: Tuesday–Saturday (left) and Sunday–Monday (right).

The Sunday - Monday segment shows somewhat higher price levels than the Tuesday - Saturday in 2025 due to poor forecasts. Nevertheless, 2026 prices are lower than the 2025 benchmarks in both segments, confirming that the price effect is not concentrated on specific days.

## 5.3 Intraday Imbalance Pattern (2026)

The heat map below reveals how average system imbalance is distributed across hours of the day and months in the 2026 data. This intraday pattern is directly relevant to trading strategy design, as persistent hour-of-day effects can inform threshold-based decision rules.

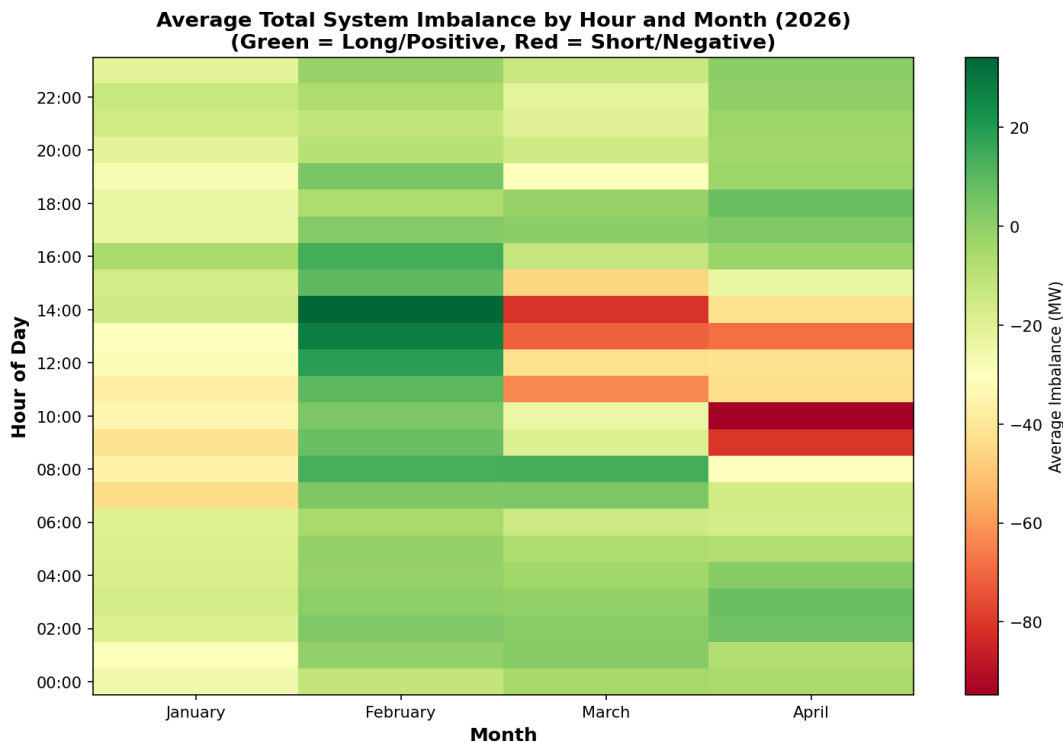


Figure 6 — Average Total System Imbalance (MW) by hour of day and month, January–April 2026. Green = net long (positive); red = net short (negative).



Several structural patterns are visible. Early morning hours (01:00 - 06:00) tend toward net long positions, likely reflecting overnight wind generation surplus relative to demand. Mid-day hours show more variable conditions, with solar generation increasingly influencing the imbalance sign as the spring months progress. The afternoon peak demand period (17:00 - 20:00) more frequently tilts to net short, calling on balancing energy to cover. These patterns provide the temporal basis for the threshold-based trading strategy evaluated in Section 6.

## 6. Imbalance Persistence Analysis

A key question for any intraday trading strategy is: if the system is imbalanced right now, how likely is it to remain imbalanced in the same direction in the near future? This section quantifies this persistence probability as a function of the imbalance threshold and the time horizon, and critically examines how this persistence changed between 2025 and 2026.

The methodology is as follows: for each 15-minute observation where the absolute imbalance exceeds a given threshold, we look forward by a fixed number of periods and check whether the imbalance direction (positive or negative) is maintained. The fraction of such events that maintain direction defines the persistence probability at that threshold.

### 6.1 Persistence in 2025 - 1.5 Hour Horizon

In 2025, with larger and more frequent imbalance episodes, the persistence probability curve is steep: relatively low magnitude thresholds are sufficient to achieve high confidence levels. The 90%, 95% and 99% confidence thresholds are reached at moderate imbalance magnitudes.

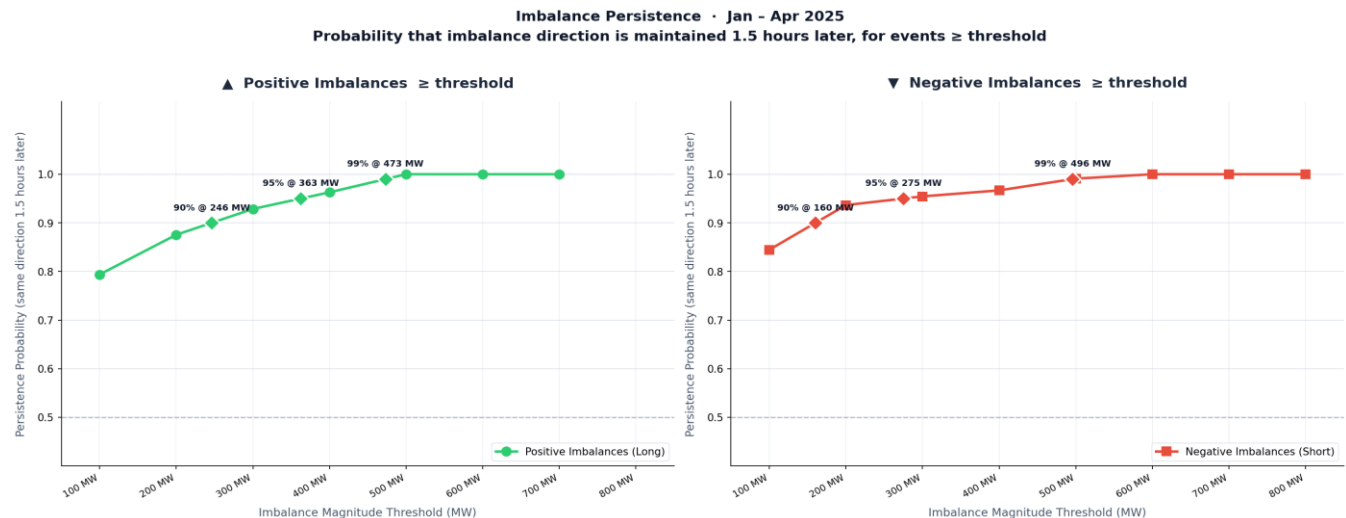


Figure 7 — Imbalance persistence probability at a 1.5-hour forward horizon, Jan–Apr 2025. Diamond markers show the magnitude thresholds at which 90%, 95% and 99% persistence probability levels are reached.

In 2025, the 90% persistence level for positive imbalances (long system) is reached at approximately 246 MW, and for negative imbalances (short system) at approximately 160 MW. These two threshold values form the basis of the intraday trading strategy in Section 7. The practical interpretation is straightforward: when the 2025 system was long by at least 246 MW or short by at least 160 MW, there was a 90% probability it would remain in the same direction 1.5 hours later.

## 6.2 Persistence in 2026 - 1.5 Hour Horizon

Repeating the same analysis for 2026 reveals a fundamental change in the persistence structure. Because the new single-price mechanism incentivises BRPs to act quickly on large imbalances, extreme imbalance events are rarer and resolve faster. As a result, the persistence probability curve flattens: the 90%, 95% and 99% confidence thresholds are no longer reached within the 100 - 800 MW range tested at the 1.5 hour horizon.

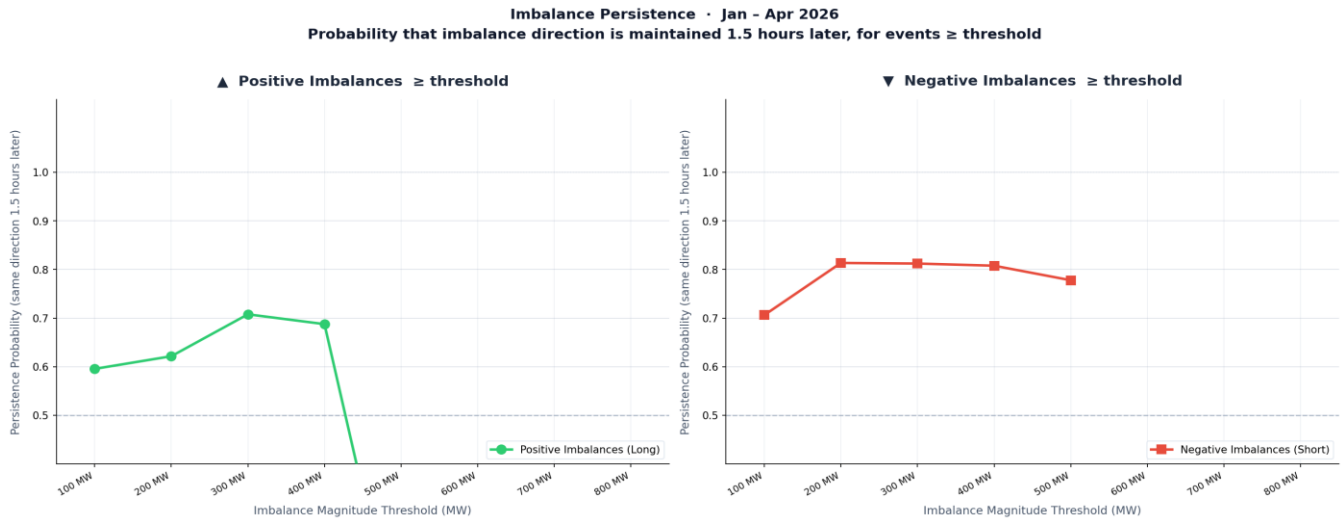


Figure 8 — Imbalance persistence probability at a 1.5-hour forward horizon, Jan–Apr 2026. The 90%, 95% and 99% probability levels are no longer reached within the observable threshold range, indicating faster imbalance resolution.

This result has a direct operational implication: the 1.5 hour decision window is no longer reliable in 2026. A trader waiting 90 minutes before acting on a large imbalance signal will find, on average, that the system has already partially corrected. The strategy must therefore use a shorter forward window.

## 6.3 Persistence in 2026 - 1 Hour Horizon

Reducing the forward horizon from 1.5 hours to 1 hour restores the observable persistence structure. At the shorter horizon, the 90% probability level is again reachable within the tested threshold range, but at higher magnitude thresholds than in 2025. The positive threshold nearly doubled (246 MW to 447 MW, +82%) while the horizon shrank by 30 minutes. The negative threshold rose significantly less than the positive threshold, from 160 to 189 MW, meaning the system still recovers slower during short imbalance. This confirms that imbalances in 2026 are less persistent overall, but sufficiently persistent over a 1 hour window at high magnitudes to support a threshold-based trading strategy.

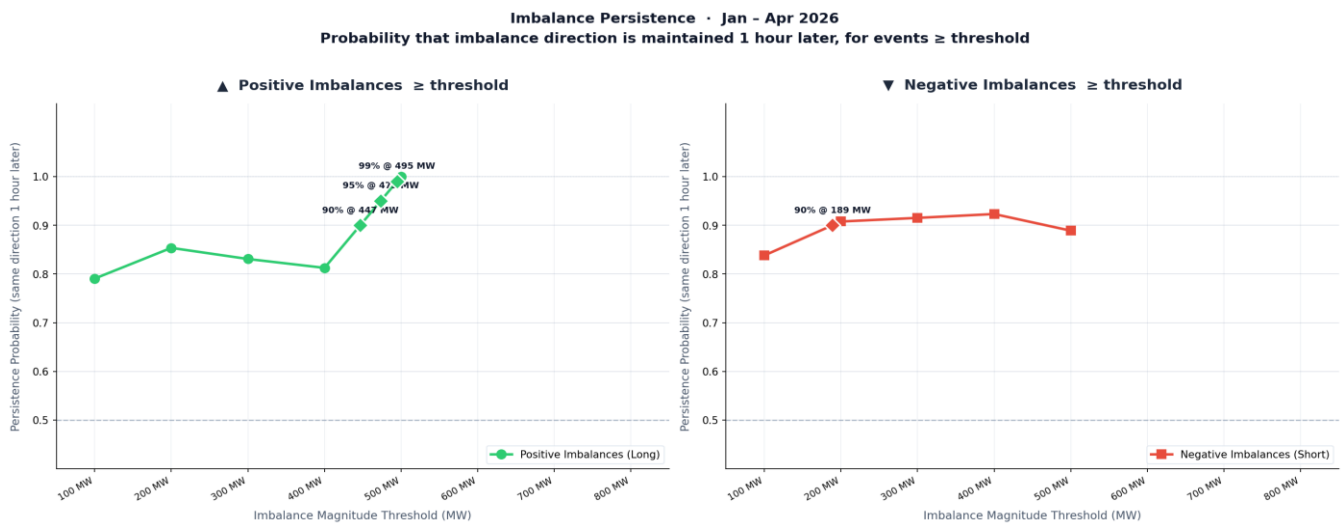


Figure 9 — Imbalance persistence probability at a 1-hour forward horizon, Jan–Apr 2026. The 90% level is now reachable, confirming that a 1-hour decision window is appropriate for 2026 conditions.

The 90% persistence thresholds derived from the 2026 one hour analysis, approximately 447 MW for positive imbalances and 189 MW for negative imbalances, are used as parameters in the trading strategy. The analysis thus establishes a data-driven rationale for both the threshold values and the reduced decision window, and highlights that the new mechanism has made intraday imbalance trading operationally more demanding: traders have less time to act and must use higher confidence thresholds.

## 7. Intraday Trading Strategy and Back-Testing

### 7.1 Strategy Logic

The back-testing framework evaluates whether historical intraday trades could have been more profitably settled via Swissgrid's imbalance price rather than at the intraday market price. The strategy is motivated by the persistence analysis: when the system imbalance exceeds the 90% persistence threshold, the position is not executed in the intraday market because it is very likely to be in the opposite direction as Swissgrid and therefore eligible for the single-price compensation.

Two conditions must simultaneously hold for a trade to qualify for Swissgrid settlement:

- **Threshold reached:** the imbalance in the preceding hour exceeds 189 MW (system short, from 2026 90% persistence level for negative imbalances) or 447 MW (system long, 2026 90% persistence level for positive imbalances). These initial thresholds represent the conservative baseline derived from the 2026 persistence analysis.
- **Opposite direction:** the trade direction is counter to the prevailing imbalance (e.g., a sell trade during a short-system episode, or a buy trade during a long-system episode). Only counter-imbalance trades can improve system balance.

When both conditions are met, the trade is re-valued at the hourly average imbalance price rather than the executed intraday price, and the PnL difference is recorded.

### 7.2 Events by Condition and Month - Initial Thresholds

Out of 2,855 total trades, only 39 (1.4%) qualify (38 are sell trades, 1 is a buy trade). Average price spread of +95 EUR/MWh, average PnL of EUR 1,174 per event. January and March show the highest counts of jointly qualifying events. February generates zero events (threshold never reached while opposite direction). April has the fewest qualifying events.

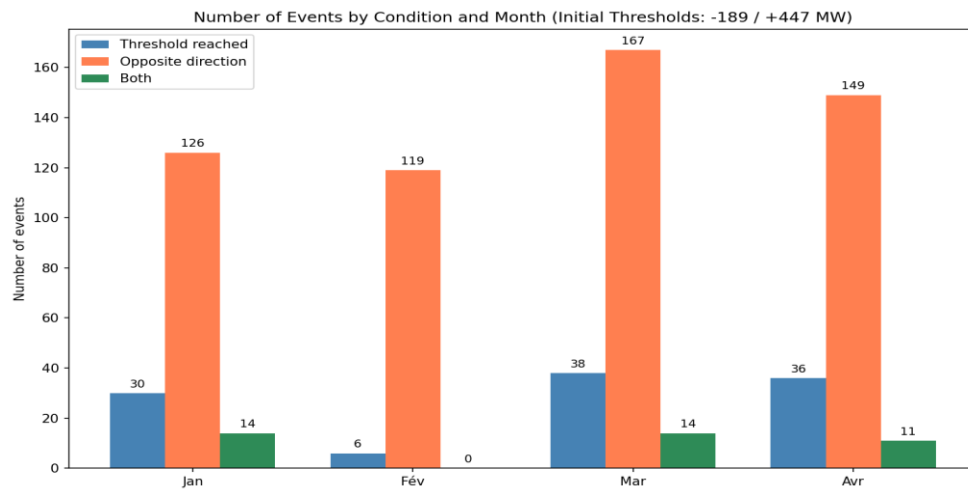


Figure 10 — Number of trades per month meeting each qualifying condition under the initial threshold parameters (-189 / +447 MW derived from 2026 persistence analysis).

### 7.3 Optimised Thresholds

Using the Nelder-Mead simplex algorithm to maximise total PnL over the four-month sample, the threshold parameters are refined. The optimised values reflect the 2026 persistence structure identified in Section 6: a slightly narrower lower bound and a wider upper bound. The shift in threshold values slightly increases the counts of qualifying events in some months.

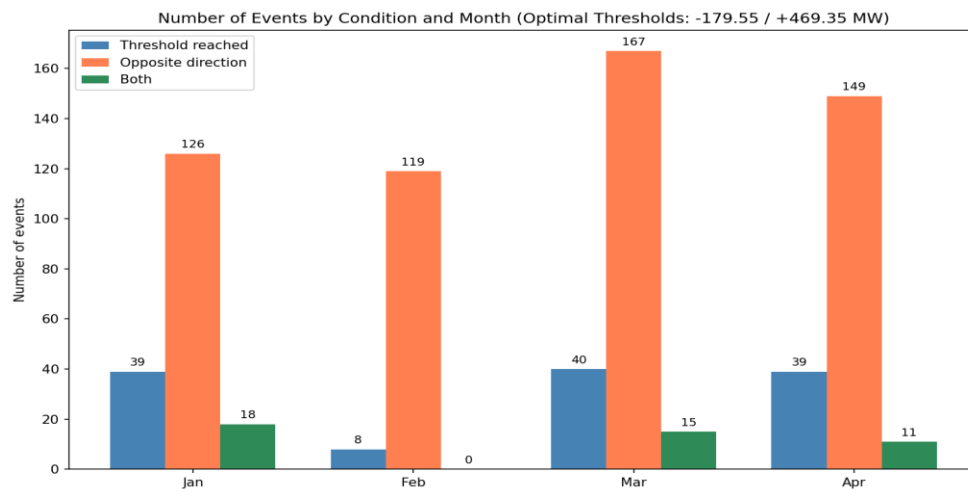


Figure 11 — Number of trades per month meeting qualifying conditions under the optimised threshold parameters (-179.55 / +469.35 MW derived from 2026 one-hour persistence analysis).

## 7.4 PnL by Month and Imbalance Direction - Threshold Strategy

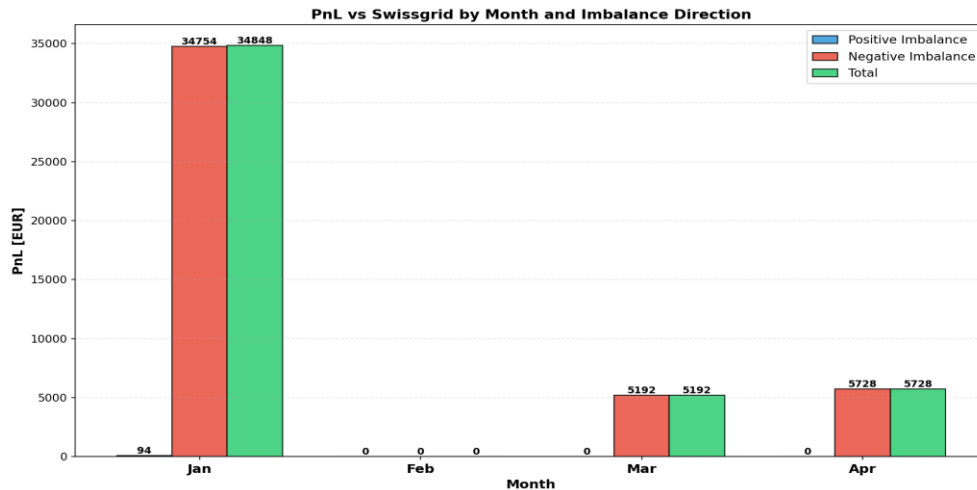


Figure 13 — PnL vs Swissgrid by month (threshold strategy), decomposed by imbalance direction: positive-imbalance trades (blue), negative-imbalance trades (red), and total (green).

Negative imbalance episodes generate the largest absolute PnL contributions across the period. In January, three consecutive sell trades on the evening of 5 January (Swissgrid prices of 689, 861, and 560 EUR/MWh) generate EUR 18,879 alone (54% of the month). In March and April, PnL is 84% lower than January due to narrower spreads and smaller average volumes.

## 8. Perfect Forecast Analysis

The perfect forecast scenario removes all operational uncertainty: the trader knows exactly (a) the direction of the system imbalance at the time of each trade, and (b) whether the Swissgrid price will be more advantageous than the intraday price. This scenario serves as a theoretical ceiling. It is unattainable in practice but provides a precise measure of the results from a perfect forecast.

The two qualifying conditions under perfect forecast are:

- EA price better: the Swissgrid imbalance price is more favourable than the executed intraday price (i.e., higher for sell trades and lower for buy trades).
- Opposite direction: the trade is counter to the prevailing imbalance direction, meaning it contributes to system balance and qualifies for the single-price compensation.

## 8.1 Events by Condition (Perfect Forecast)

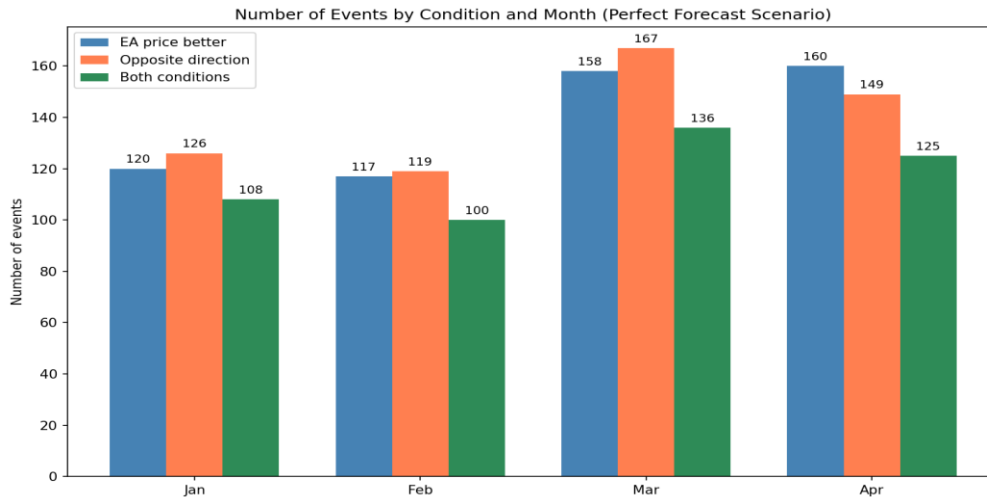


Figure 14 — Number of trades per month meeting each condition under the perfect forecast scenario: EA price better (blue), opposite direction (coral), and both conditions jointly (green).

Out of 2,855 total trades, 469 (16.4%) qualify. This is a massive increase. The jointly qualifying counts is highest in March and April.

## 8.2 PnL Decomposition under Perfect Forecast

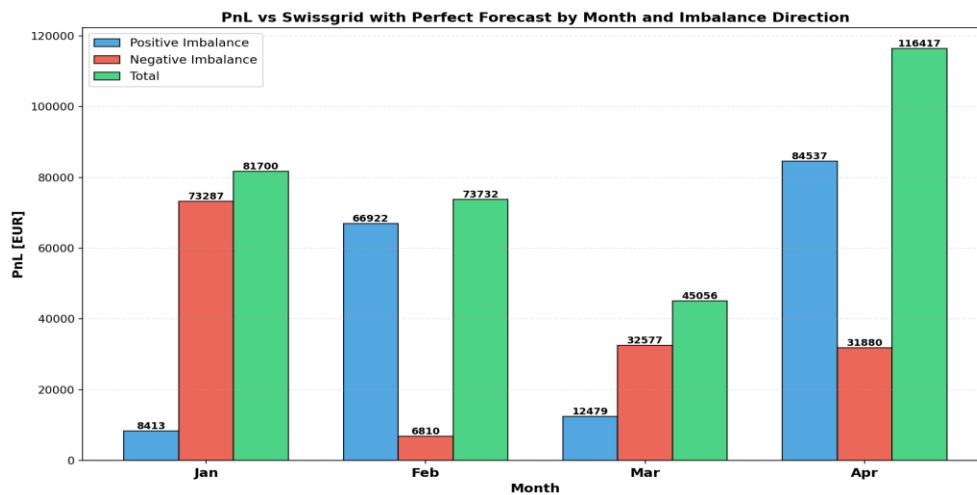


Figure 15 — PnL vs Swissgrid under the perfect forecast scenario by month and imbalance direction: positive-imbalance contributions (blue), negative-imbalance contributions (red), and monthly total (green).

Under perfect forecast, total PnL reaches EUR 316,904 over the four-month period compared to EUR 45,768 under the initial threshold strategy and EUR 50,455 under the optimised thresholds. All months and both imbalance directions contribute positively, with April generating the largest gains due to both the higher frequency of extreme imbalance events and the wider spread between intraday and imbalance prices in that month. The gap between the optimised threshold strategy and the perfect forecast ceiling amounts to EUR 266,449 over four months, representing the theoretical upside achievable through a high-quality short-term imbalance price prediction model. This quantifies the economic incentive for investing in machine-learning or statistical forecasting tools that can predict the sign and relative level of the price for the next hour.

## 9. Key Findings

### On system imbalance:

- The introduction of the single-price mechanism in January 2026 is associated with a 39.1% reduction in average absolute system imbalance, from 129.1 MW to 78.6 MW per 15-minute interval, comparing January - April 2025 to January–April 2026.
- The reduction is statistically significant at the 0.1% level (Mann-Whitney U,  $p < 0.001$ ) for overall, positive, and negative imbalance episodes individually.
- The improvement is consistent across both day-group segments (Tuesday - Saturday and Sunday - Monday), confirming it is not driven by a particular day type.
- Average imbalance prices declined alongside volumes, consistent with lower merit-order activation requirements under reduced imbalance conditions.

### On imbalance persistence:

- In 2025, imbalances above ~189 MW (positive) and ~447 MW (negative) persisted in the same direction with 90% probability over a 1.5-hour horizon.
- In 2026, the 90% persistence threshold is no longer reached within the same magnitude range at the 1.5-hour horizon, indicating faster imbalance resolution driven by the new financial incentives.
- Reducing the horizon to 1 hour restores the observable persistence structure in 2026, but at higher threshold magnitudes (approx. -179.55 MW and +469.35 MW). Traders therefore have less time to decide and must act on more extreme signals.

### On the intraday trading strategy:

- A threshold-based strategy using the 2026 90% persistence thresholds (-189 / +447 MW) generates a total PnL of EUR 45,768 vs Swissgrid settlement prices over January - April 2026.
- Optimising the thresholds to the 2026 one-hour persistence values (-179.55 / +469.35 MW) improves total PnL to EUR 50,455, a gain of EUR 4,687 (+10.2%).
- The theoretical perfect-forecast upper bound, assuming exact knowledge of both imbalance direction and relative price advantage, reaches EUR 316,904 over the same period, a gap of EUR 266,449 above the optimised threshold strategy.
- The large gap between optimised threshold and perfect forecast performance identifies accurate short-term imbalance price forecasting as the primary lever for commercial improvement. This motivates investment in predictive models at the gate-closure horizon.

### On limitations:

- The analysis covers only January - April in both years and may not generalise to other seasons where renewable production patterns and demand profiles differ substantially.
- The arithmetic mean was used as the central tendency metric overstating the typical 15 minutes interval compared to the median (though this bias is identical in both years).
- Threshold optimisation is performed in-sample. Out-of-sample validation via walk-forward testing is required before production deployment.
- The back-test assumes the full intraday trade volume can be offset against the system imbalance, without verifying that the traded volume does not exceed the prevailing imbalance magnitude.

- One hour time horizon may be too short in practice to take a decision as gate close 5 minutes before delivery on the Swiss intraday market. Also, liquidity near time closure may be lower and subject to high price volatility. Furthermore, Swissgrid doesn't provide real time imbalance data.
- A systematic risk remains: if multiple BRPs simultaneously adopt counter-imbalance strategies based on the same public signal, the imbalance may shift in the same direction as all of them, converting expected gains into penalties.

## 10. Conclusion

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This study provides quantitative evidence that Swissgrid's transition to a single-price imbalance settlement mechanism at the start of 2026 has materially improved Swiss power system balance in its first four months of operation. The mechanism appears to have generated exactly the financial incentives its designers intended: balance responsible parties have adjusted their trading and optimisation behaviour, reducing both the frequency and magnitude of extreme imbalance events. However, the same mechanism that reduces imbalance also makes it harder to exploit: faster resolution means traders have less time to act and must rely on more extreme signals. The back-testing analysis demonstrates that even a simple lagged threshold rule can generate meaningful positive PnL, but the wide gap to the perfect-forecast ceiling points to forecast quality as the next frontier. Future work should focus on developing short-horizon machine-learning models for price and imbalance prediction, validated out-of-sample and stress-tested against scenarios where multiple BRPs simultaneously target counter-imbalance positions.

## References

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- [1] <https://www.swissgrid.ch/en/home/newsroom/newsfeed/20250328-01.html>  
[2] <https://www.swissgrid.ch/fr/home/customers/topics/bgm/balance-energy.html>

## Statement

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